

DAY 6 – Python Data Structures (Lists, Tuples, Sets & Dictionaries)

This is one of the most important topics for interviews, DSA, AI/ML preprocessing, and everyday Python programming.



What You'll Learn Today

1. Lists

Lists are ordered, mutable collections.



```
fruits = ["Apple", "Banana", "Mango"]  
print(fruits) print(fruits[0])
```

Common Methods



```
fruits.append("Orange")  
fruits.remove("Banana") fruits.insert(1,  
"Grapes") fruits.sort() print(fruits)
```

2. Tuples

Tuples are ordered but immutable.



```
coordinates = (10, 20)  
print(coordinates)
```

Why use tuples?

Faster

Cannot be changed accidentally

Used for fixed values

3. Sets

Sets store unique values.

```
numbers = {1,2,3,3,4,4} print(numbers)
```



Output

```
{1,2,3,4}
```



Methods



```
numbers.add(10) numbers.remove(2)
```

Useful for

Removing duplicates

Fast searching

4. Dictionaries ★★★★★

Most important data structure.

Stores key-value pairs.



```
student = { "name":"Utkarsh", "age":20,  
"course":"AI ML" } print(student)
```

Access



```
print(student["name"])
```

Update



```
student["age"] = 21
```

Add



```
student["college"] = "ABC College"
```

Loop



```
for key,value in student.items():  
    print(key,value)
```

Mini Project

Student Database



```
students = { "Utkarsh":90, "Rahul":82,  
"Ankit":95 } for name,marks in  
students.items(): print(f"{name}  
scored {marks}")
```

Output



```
Utkarsh scored 90 Rahul scored 82  
Ankit scored 95
```

Practice Questions

Easy

 Create a list of five programming languages.

 Print the third element.

 Add "Go" to the list.

 Remove "Java".

Medium

Create a dictionary



```
Name Age Branch CGPA
```

Print all values.

Challenge

Take five numbers from the user.

Store them in a list.

Print

Maximum

Average

Interview Questions

Difference between List and Tuple?

List	Tuple
Mutable	Immutable
[]	()
Slower	Faster

Difference between Set and List?

List	Set
Ordered	Unordered
Duplicate Allowed	Unique Values

Difference between Dictionary and List?

Dictionary	List
Key-Value Pair	Index Based

Real-Life AI/ML Usage

Lists → Store datasets, predictions, feature values.

Tuples → Fixed coordinates, image dimensions.

Sets → Remove duplicate records before training.

Dictionaries → Model parameters, JSON data, API responses, label mappings.

Mini Project: Student Management System (Python DSA)



```
# Student Management System
students = {} while True:
print("\n===== Student Management
=====") print("1. Add Student")
print("2. View Students") print("3.
Search Student") print("4. Remove
Student") print("5. Exit") choice =
input("Enter your choice: ") if choice ==
"1": name = input("Enter Name: ") age =
int(input("Enter Age: ")) marks =
float(input("Enter Marks: "))
students[name] = { "Age": age, "Marks":
marks } print("✅ Student Added
Successfully!") elif choice == "2": if not
students: print("No students found.")
else: print("\nStudent Records")
print("-----") for name,
details in students.items():
print(f"Name : {name}") print(f"Age :
{details['Age']}") print(f"Marks:
{details['Marks']}")
print("-----") elif choice ==
```

```
"3": name = input("Enter student name:
") if name in students:
print(students[name]) else:
print("Student Not Found!") elif choice
== "4": name = input("Enter student
name: ") if name in students: del
students[name] print("Student
Removed!") else: print("Student Not
Found!") elif choice == "5": print("Thank
You!") break else: print("Invalid Choice")
```

DSA Concepts Used

Dictionary

```
students = {}
```



✓ Nested Dictionary



```
students[name] = { "Age": age, "Marks":  
marks }
```

✓ Search



```
if name in students:
```

✓ Traversal



```
for name, details in students.items():
```



Delete



```
del students[name]
```

Bonus Challenge

Add:



Average Marks Topper Sort Students
Update Marks Count Students

These features make the project more complete and prepare you for real-world CRUD applications.

